

Power BI DAX Lab (Accounting) — Print-Ready Student Handout

Dataset: PowerBI_DAX_Accounting_Student.xlsx

Part A — Import + Model Setup (10–15 min)

1) Import tables: Fact_Transactions, Dim_Account, Dim_Department, Dim_Entity, Calendar_Reference.

2) Create relationships (Star Schema):

From (Fact)	To (Dim)
Fact_Transactions[AccountNo]	Dim_Account[AccountNo]
Fact_Transactions[DeptCode]	Dim_Department[DeptCode]
Fact_Transactions[EntityID]	Dim_Entity[EntityID]
Fact_Transactions[TransDate]	Calendar_Reference[Date]

3) Mark Date Table: Calendar_Reference → Mark as date table → Date.

Part B — Required Measures (copy/paste DAX)

Net Amount

Net Amount := SUM(Fact_Transactions[Net])

Total Debit / Credit

Total Debit := SUM(Fact_Transactions[Debit])

Total Credit := SUM(Fact_Transactions[Credit])

Transaction Count

Transaction Count := COUNTROWS(Fact_Transactions)

Distinct Entities

Distinct Entities := DISTINCTCOUNT(Dim_Entity[EntityID])

Revenue / Expense

Revenue := CALCULATE([Net Amount], Dim_Account[AccountType] = "Revenue")

Expense := CALCULATE([Net Amount], Dim_Account[AccountType] = "Expense")

Net Income

Net Income := [Revenue] - [Expense]

Profit Margin %

Profit Margin % := DIVIDE([Net Income], [Revenue])

Posted Net

Posted Net := CALCULATE([Net Amount], Fact_Transactions[Status] = "Posted")

YTD Net

YTD Net := TOTALYTD([Net Amount], Calendar_Reference[Date])

Net PY

Net PY := CALCULATE([Net Amount], SAMEPERIODLASTYEAR(Calendar_Reference[Date]))

YoY Change / %

YoY Change := [Net Amount] - [Net PY]
YoY % := DIVIDE([YoY Change], [Net PY])

Material Net (FILTER)

Material Net (>=5000) := CALCULATE([Net Amount], FILTER(Fact_Transactions, ABS(Fact_Transactions[Net]) >= 5000))

Missing Dept Count (FILTER)

Missing Dept Count := COUNTROWS(FILTER(Fact_Transactions, Fact_Transactions[DeptCode] = ""))

Extended Amount (SUMX)

Extended Amount (SUMX) := SUMX(Fact_Transactions, Fact_Transactions[Quantity] * Fact_Transactions[UnitPrice])

% of Total (ALL)

% of Total by Account := DIVIDE([Net Amount], CALCULATE([Net Amount], ALL(Dim_Account)))

Vendor Rank / Top 5

Vendor Rank := RANKX(ALL(Dim_Entity[EntityName]), [Net Amount], , DESC)
Top 5 Vendor Spend := CALCULATE([Net Amount], TOPN(5, ALL(Dim_Entity[EntityName]), [Net Amount], DESC))

Part C — Required Visuals

- 1) Monthly Trend (Line/Column): Axis = Calendar_Reference[YearMonth], Values = [Net Amount], slicer = Status.
- 2) Income Statement Matrix: Rows = Dim_Account[AccountType], Columns = Calendar_Reference[YearMonth], Values = [Net Amount].
- 3) Exception Cards: [Material Net (>=5000)], [Missing Dept Count], [Transaction Count].
- 4) Top Vendors Bar Chart: Axis = Dim_Entity[EntityName], Values = [Net Amount], show Top 5.

Part D — Student Questions (submit short answers)

- Q1) What is Net Income for the full period?
- Q2) Why is [Posted Net] different from [Net Amount]?
- Q3) How many transactions are missing DeptCode? Why is that a reporting risk?
- Q4) Why do we need SUMX to compute Extended Amount (Qty × UnitPrice)?
- Q5) Which vendor has the highest net impact? (Use Vendor Rank / Top 5.)

Deliverables: Screenshot of relationships, screenshots of visuals, and DAX measures list.